Chapter 8, Problem 10P

Problem

The differential equation that describes the current in an *RLC* network is

$$3\frac{d^2i}{dt^2} + 15\frac{di}{dt} + 12i = 0$$

Given that *i*(0) = 0, *di*(0)/*dt* = 6 mA/s, obtain *i*(*t*).

Step-by-step solution

Step 1 of 5

Consider the differential equation for the current.

$$3\frac{d^2i}{dt^2} + 15\frac{di}{dt} + 12i = 0$$

$$\frac{d^2i}{dt^2} + 5\frac{di}{dt} + 4i = 0$$

Treat $\frac{d}{dt}$ as s.

$$s^2 i + 5si + 4i = 0$$

$$\left(s^2 + 5s + 4\right)i = 0$$

Step 2 of 5

Consider the characteristic equation.

$$s^2 + 5s + 4 = 0$$

$$(s+4)(s+1) = 0$$

The roots of the characteristic equation are $s_1 = -4$ and $s_2 = -1$.

Step 3 of 5

The roots are negative and real. Hence, the equation exhibits the over damped response.

Consider the expression for the current $i(t)$.

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute -4 for s_1 and -1 for s_2.

$$i(t) = A_1 e^{-4t} + A_2 e^{-1t} \text{ ...... (1)}$$

Substitute 0 for t.

$$i(0) = A_1 e^{-4(0)} + A_2 e^{-1(0)}$$

Substitute 0 for $i(0)$.

$$0 = A_1 e^{(0)} + A_2 e^{(0)}$$

$$0 = A_1(1) + A_2(1)$$

$$0 = A_1 + A_2$$

$$A_1 = -A_2$$

Recall equation (1).

$$i(t) = A_1 e^{-4t} + A_2 e^{-1t}$$

Apply derivative with respect to t.

$$\begin{aligned}\frac{di(t)}{dt} &= A_1 \frac{de^{-4t}}{dt} + A_2 \frac{de^{-1t}}{dt} \\ &= -4A_1 e^{-4t} - 1A_2 e^{-1t}\end{aligned}$$

Substitute 0 for t.

$$\begin{aligned}\frac{di(0)}{dt} &= -4A_1 e^{-4(0)} - A_2 e^{-1(0)} \\ &= -4A_1 e^{(0)} - A_2 e^{(0)} \\ &= -4A_1(1) - A_2(1) \\ &= -4A_1 - A_2\end{aligned}$$

Substitute 6 mA/s for $\frac{di(0)}{dt}$ and $-A_2$ for A_1.

$$6\text{ m} = -4(-A_2) - A_2$$

$$6\text{ m} = 4A_2 - A_2$$

$$3A_2 = 6\text{ m}$$

$$A_2 = \frac{6\text{ m}}{3}$$

$$A_2 = 2\text{ m}$$

Therefore, the value of the constant A_2 is 2 m.

The value of the constant A_1 is -2 m since $A_1 = -A_2$.

Recall equation (1).

$$i(t) = A_1 e^{-4t} + A_2 e^{-1t}$$

Substitute -2 m for A_1 and 2 m for A_2.

$$\begin{aligned}i(t) &= (-2\text{ m})e^{-4t} + (2\text{ m})e^{-1t} \\ &= 2\left(e^{-1t} - e^{-4t}\right)\text{ mA}\end{aligned}$$

Therefore, the value of the current $i(t)$ is $\boxed{2\left(e^{-1t} - e^{-4t}\right)\text{ mA}}$.